

Derbyshire Shared Care Pathology Guidelines

Dyslipidaemia

This guideline applies to patients with significantly abnormal lipid profiles, which may be primary (genetic), secondary to other diseases and drugs, or a combination of both. It is intended to give some general information and suggestions for initial assessment and treatment for non-specialists. It is not intended to be fully comprehensive or a replacement for discussion of individual cases with a lipid specialist.

There is a great difference between managing average lipid levels in patients at high risk of CVD and those with severe dyslipidaemias. Derbyshire guidelines for adult lipid modification therapy can be found here: [Lipid modification therapy in non-fh.pdf \(derbyshiremedicinesmanagement.nhs.uk\)](https://www.derbyshiremedicinesmanagement.nhs.uk/Lipid_modification_therapy_in_non-fh.pdf)

Samples for lipid measurements

A non-fasting sample for lipid profile is appropriate in the vast majority of patients.

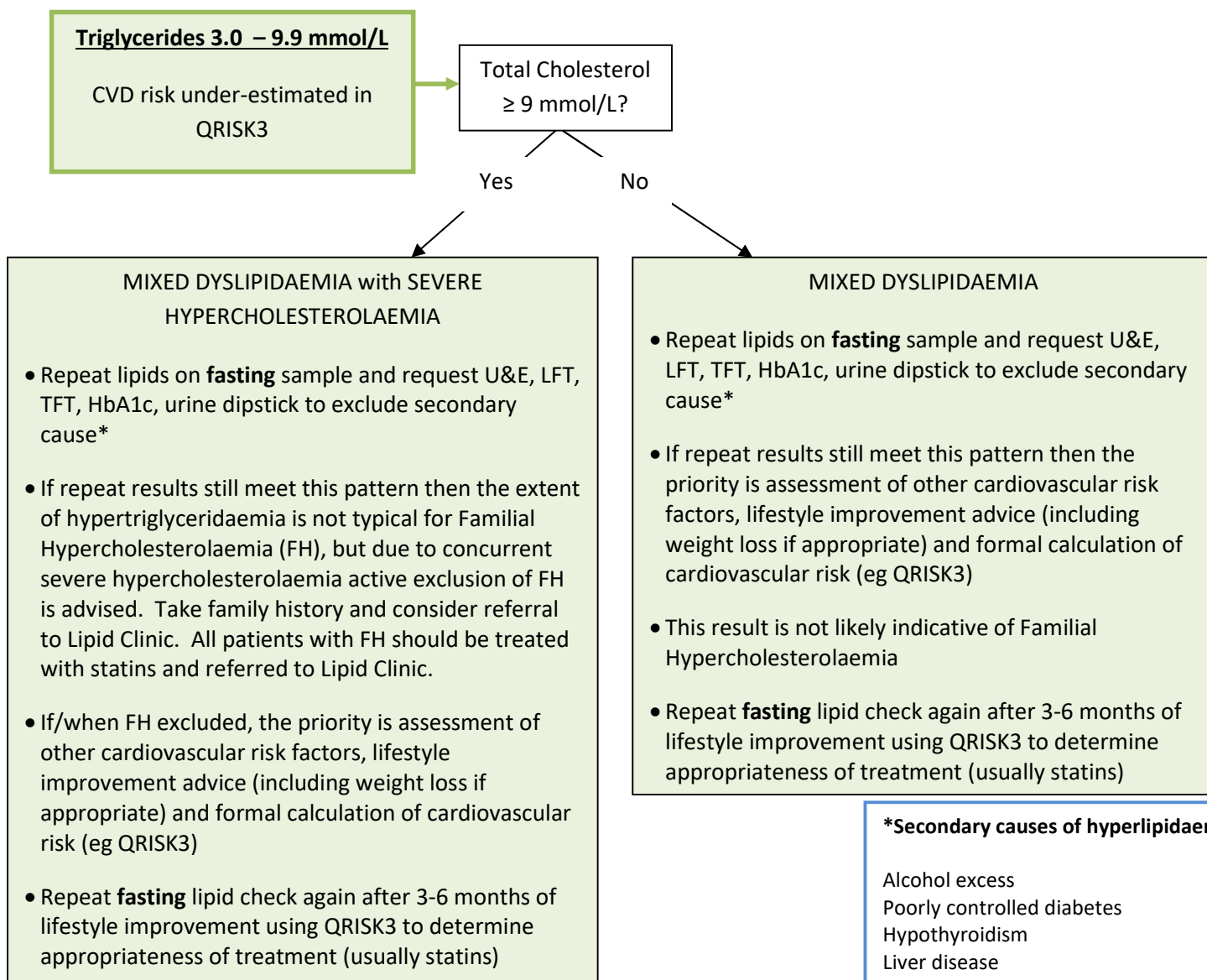
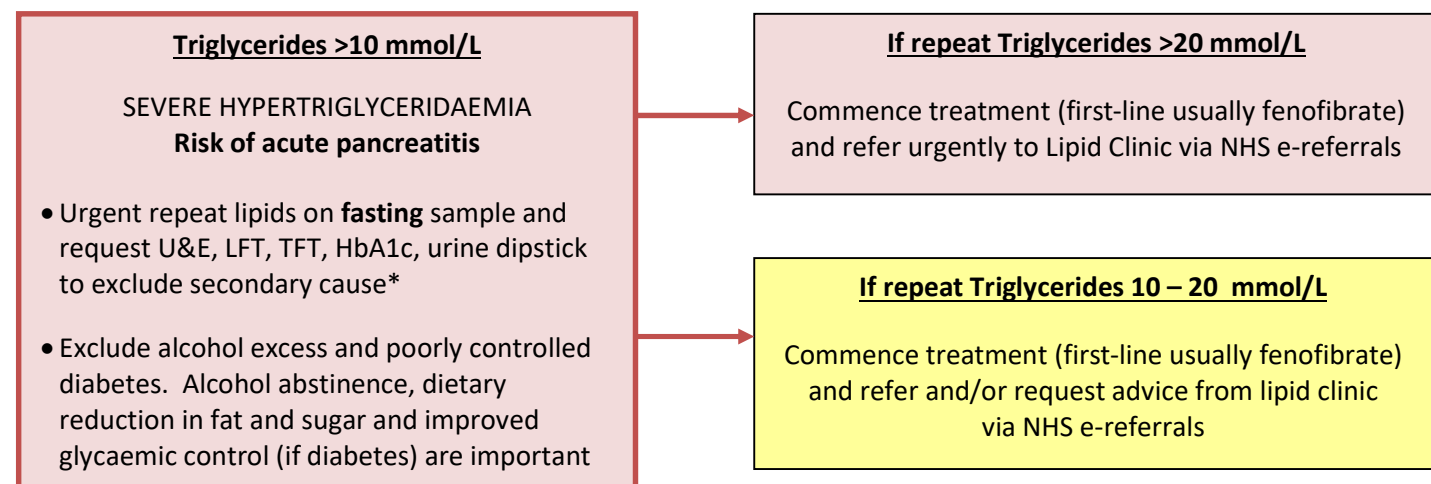
A fasting sample may be required for follow up testing if directed in the flow charts below.

Interpreting the lipid report

Abnormal lipid profiles fall into the following broad categories. Please refer to the flowcharts on the following pages:

- Severe hypertriglyceridaemia
- Severe hypercholesterolaemia
 - Due to LDL
 - Due to HDL
 - Due to both
- Mixed hyperlipidaemia
- Low HDL

Investigation and Management of Hypertriglyceridaemia



*Secondary causes of hyperlipidaemia

Alcohol excess
Poorly controlled diabetes
Hypothyroidism
Liver disease
Nephrotic syndrome

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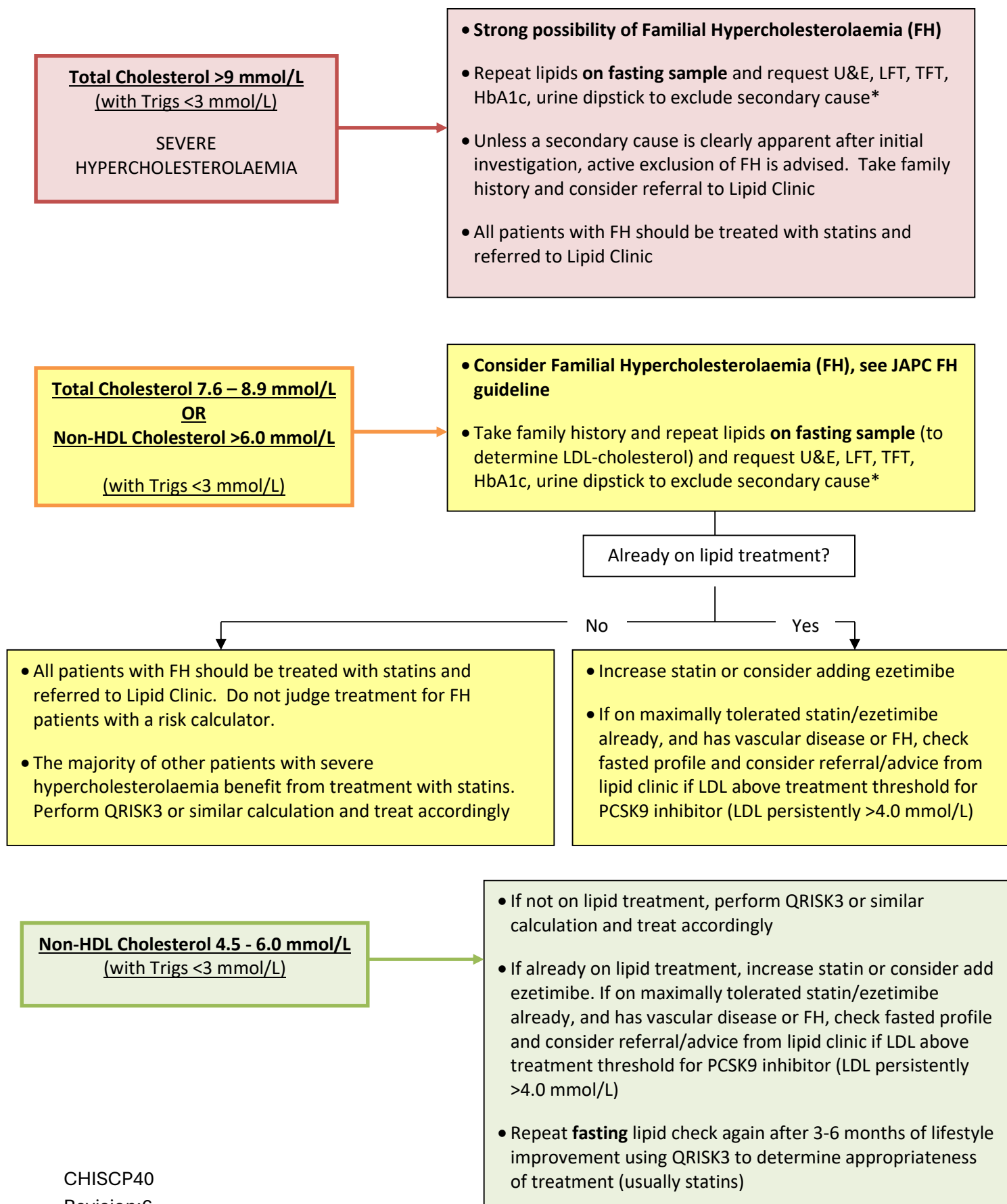
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Page 2 of 6

Investigation and Management of Hypercholesterolaemia



Lipid Interpretation Background Information

Severe hypertriglyceridaemia (>10 mmol/L)

NB: Cholesterol is likely to be secondarily elevated as the lipoprotein particles also contain cholesterol.

Risk: Acute pancreatitis, CVD risk uncertain

Causes: Genetic predisposition, plus diabetes (especially if poorly controlled), obesity, alcohol, diet high in sugar and/or total fat. Drugs: steroids, isotretinoin, oral oestrogens

Clinical features: History of abdominal pain, rashes (eruptive xanthomata)

Treatment: Underlying cause (where present) plus fibrate and/or fish oil. Statins alone are very unlikely to be effective. Discuss with or refer to lipid clinic if uncertain.

With triglycerides above 10 mmol/L there is an increasing risk of acute pancreatitis.

Triglycerides may be >50 or even >100 mmol/L in some individuals. The sample is likely to be reported as lipaemic by the lab. It is important to recognise that severe hypertriglyceridaemia is abnormal whether or not the sample was taken fasting.

Triglyceride-induced pancreatitis may occur with normal serum amylase, due to interference with its measurement. Urine amylase, however, is unaffected and must be measured if the serum amylase is normal.

Severe hypercholesterolaemia

Raised LDL

Risk: Cardiovascular

Causes: Genetic (particularly familial hypercholesterolaemia). May be secondary to hypothyroidism, nephrotic syndrome, cholestasis

Look for: Tendon xanthoma, eyelid xanthelasma, corneal arcus

Treatment: Statin (usually high intensity). Current 1st choice is atorvastatin. Dose should be the lowest required to achieve adequate lipid control (eg 50% fall in LDL). Ezetimibe may be considered in accordance with NICE technology appraisal guidance TA132.

Familial hypercholesterolaemia (FH) should be considered in adult patients with LDL >5 mmol/L and is more likely the higher LDL. However, most patients with LDL 5-7 mmol/L will not have FH, especially if there is a history of previous lower results in the laboratory record. Conversely, relatives of patients with known FH may have LDL <5 mmol/L and still have the condition. Genetic testing followed by cascade screening is recommended by NICE, but availability of this across England remains low. Derbyshire FH guidelines (Identification and Management of Familial Hypercholesterolaemia) can be found here:

[Identification and management of familial hypercholesterolaemia \(FH\)](https://derbyshiremedicinesmanagement.nhs.uk/identification-and-management-of-familial-hypercholesterolaemia-fh/)
([derbyshiremedicinesmanagement.nhs.uk](https://derbyshiremedicinesmanagement.nhs.uk/derbyshiremedicinesmanagement.nhs.uk))

Raised HDL

Risk: None, when occurring naturally

Causes: Mainly genetic, but also alcohol, oestrogens (HRT and combined OCP), some anticonvulsants (particularly phenytoin), type 1 diabetes (insulin sensitive phenotype)

Treatment: Not usually required, unless QRISK3 score is >10%

Raised HDL has been shown to be associated with lower cardiovascular events in epidemiological studies. It is not clear whether HDL raised by drugs or alcohol is as protective as high levels occurring naturally. In these patients, look at the LDL or non-HDL cholesterol before deciding whether treatment is needed. HDL may be above 3 mmol/L in some individuals and exceptionally up to 5 mmol/L.

Mixed hyperlipidaemia

Causes: Type III hyperlipoproteinaemia, other genetic predisposition, obesity, diet, alcohol

Risk: Cardiovascular

Treatment: Statin first line: current first choice is atorvastatin. Type III hyperlipidaemia may respond particularly well to fibrates.

Clinical features: Palmar xanthoma (rare, but indicate type III)

Mixed hyperlipidaemias are usually a combination of a genetic predisposition, plus an aggravating factor as with severe hypertriglyceridaemia. However, if triglycerides are <10 mmol/L the risk of acute pancreatitis is not significantly raised. This type of pattern is anecdotally seen in patients with early onset of cardiovascular disease. There is often underlying insulin resistance and a risk of future type 2 diabetes (see low HDL below).

Low HDL

Risk: Cardiovascular

Causes: Genetic predisposition, insulin resistance, inflammatory diseases, smoking

Treatment: Statin, if indicated by QRISK3. Current 1st choice is atorvastatin.

Low HDL may cause very raised Cholesterol:HDL ratio (eg 30) in some individuals and is often seen with mild to moderately raised triglycerides (<5 mmol/L). It is often due to insulin resistance and may pre-date type 2 diabetes by a number of years. It may be associated with other features of the insulin resistance/ metabolic syndrome eg fatty liver disease, polycystic ovaries. HDL can be difficult to raise significantly by lifestyle measures. Raising HDL through drug treatment with fibrates and nicotinic acid has not been shown to reduce CVD events, however. Lowering total and LDL cholesterol with statins is the currently favoured approach, but there is scant evidence to guide targets.

Lipid clinic referrals

Referral to a lipid clinic is available where there is uncertainty regarding diagnosis and for management advice, including the use of non-statin drugs and combination treatment. Genetic testing and cascade screening for familial hypercholesterolaemia may become

more widely available and patients who have been assessed in a lipid clinic will be the first to access these.

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